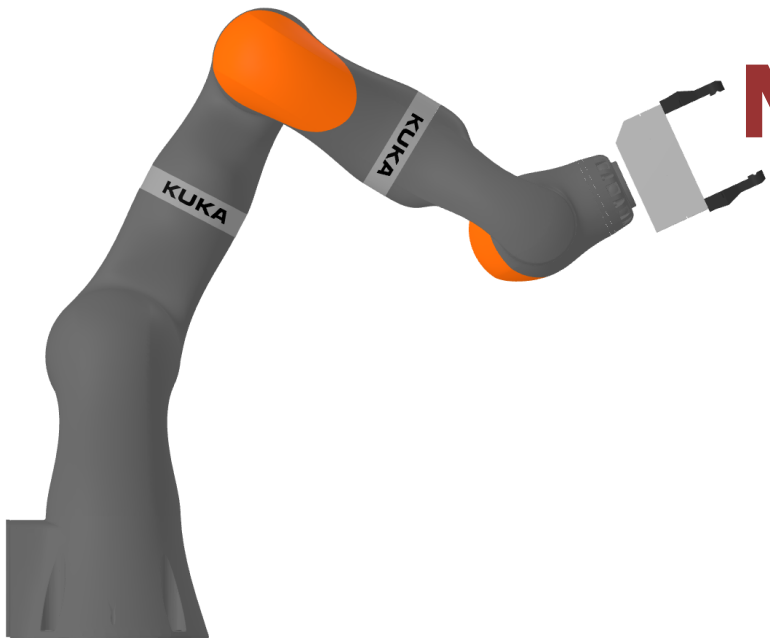
Introduction: Anatomy of a Manipulation System



MIT 6.4210/2: Robotic Manipulation

Fall 2024, Lecture 1

Massachusetts Institute of Technology

Course Info

https://manipulation.csail.mit.edu/Fall2024/index.html

Website: http://manipulation.mit.edu

6.4210 is a CI-M (Communication Intensive in the Major)

Communications Instructors from Comparative Media Studies/Writing

Recitations Fridays afternoons (starting Sept 13)

Project-related (CI) assignments for 6.4210 students

1. Journal club presentation *(3 pts)*
2. Project preproposal *(1 pt)*
3. Project proposal final revision *(4 pts)*
4. Project final report *(10 pts)*
5. Project final video *(3 pts)*
6. Peer reviews *(2 pts)*
7. Writing conferences *(2 pt)*
8. CI-M participation & pre-writing tasks *(3 pts)*

Logistics

- Students: Make sure you're on Piazza
  - CI-M (6.4210) material on Canvas.
- Review the course guidelines.
- Read/comment on the lecture notes
-
- ~ weekly problem sets throughout the class, due Tue.
  - Problem Set 1 released today (on the course calendar).
- Final project.

from the course website:

Course Information

Class Time and Location

Fall semester (September - December, 2024)
Lectures: Monday, Wednesday 9:30 -11:00 am in 45-230
Recitations are only for the undergrad version (6.4210):

- R1: Friday at 1pm in 4-159,
- R2: Friday at 2pm in 4-149, or
- R3: Friday at 1pm in 4-149.

See class schedule for complete details.

Office Hours

TBD

Class Forums

Forums are on Piazza (sign up with your @mit.edu email address).

For 6.4210, communication on CI-M material will primarily happen through Canvas.

We will use Gradescope for problem set submission and grading.

Grading Policy

Late assignments will be penalized 10% every 24 hours. Additionally, we will grant a one time (one pset) extension of up to one week without penalty. This will automatically be applied at the end of the term to whichever pset will earn you the most points; there is no need to request it. All psets will be be assigned equal weight when determining final grade.

6.4210 grade distribution
Assignments: 40%
Journal club/peer reviews: 5%
CI-M participation, pre-writing tasks & writing conferences: 5%
Project proposal: 10% (5% tech grade + 5% CI grade)
Project report: 30% (20% tech grade + 10% CI grade)
Project video: 10% (7% tech grade + 3% CI grade)

6.4212 grade distribution
Assignments: 40%
Project proposal: 5%
Project updates: 10%
Project report: 35%
Project video: 10%

Policy on Generative AI: Generative AI is a resource like any other. You are free to use it, but should always cite your resources.

https://manipulation.csail.mit.edu/intro.html

https://people.csail.mit.edu/russt/uploads/yellow_spot.html

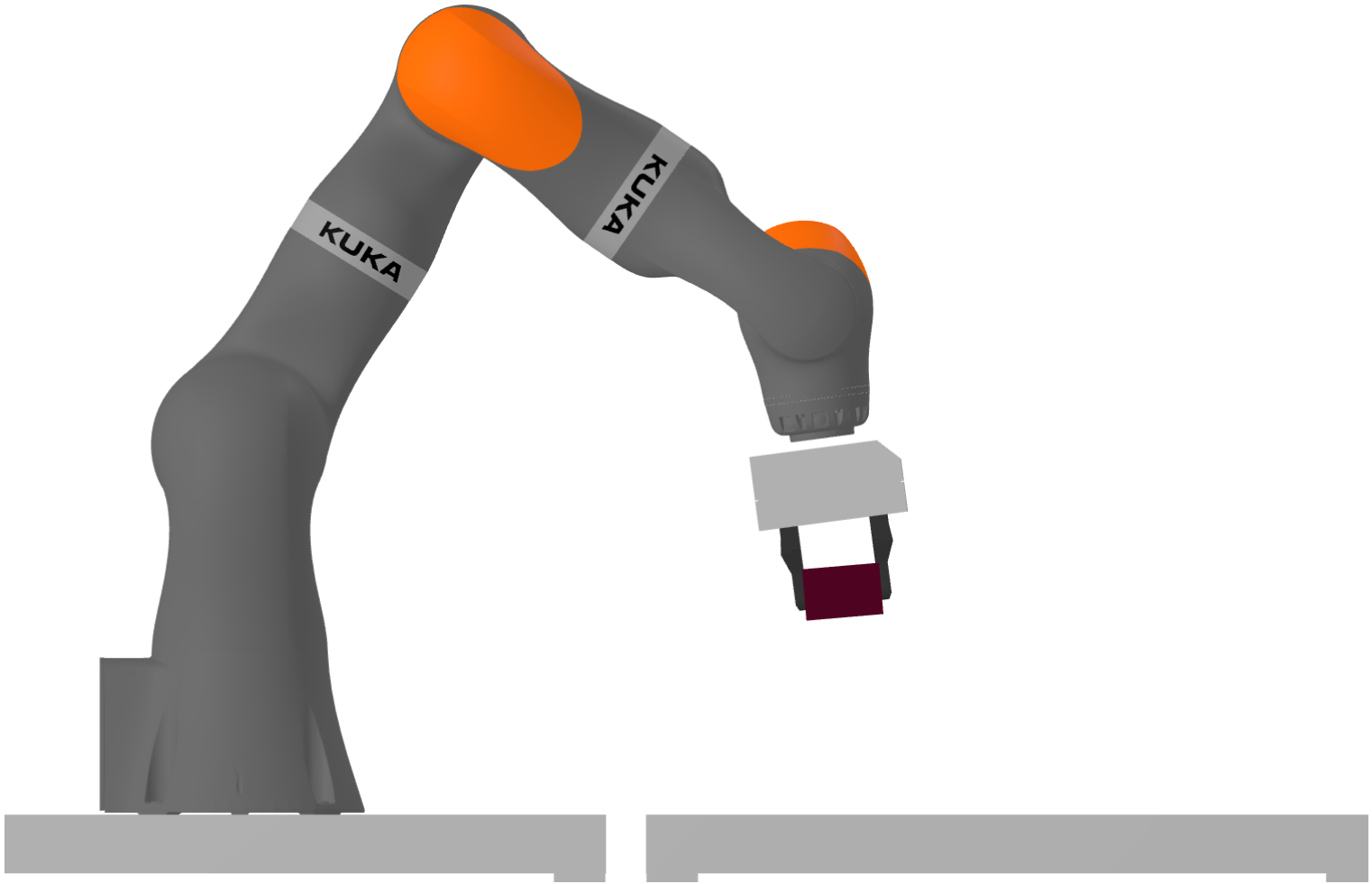
Today

- What do we mean by "***manipulation***"? (with examples!)
- The systems theory perspective
- Anatomy of a manipulation system
- Goals for the course

Matt Mason says

- *Definition 1 (etymological).* Manipulation refers to the activities performed by hands.
- …
- *Definition 5.* Manipulation refers to an agent's control of its environment through selective contact.

Matthew T. Mason. Toward Robotic Manipulation. *Annual Review of Control, Robotics, and Autonomous Systems*, 1:1-28, 2018.


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https://www.youtube.com/embed/hxsWeVtb-JQ?enablejsapi=1

https://www.youtube.com/embed/1RbaIo4VdbA? enablejsapi=1&start=112&mute=1

Autonomous "Open-world" Manipulation

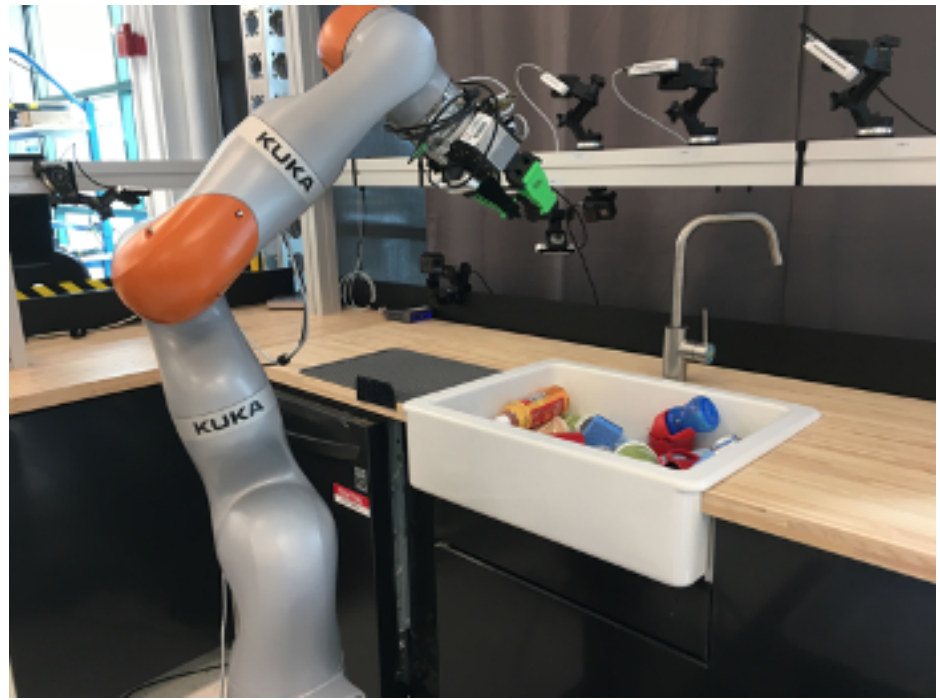
- *Definition 5.* Manipulation refers to an agent's control of its environment through selective contact.

- *Even broader:* "Open-world" manipulation requires:
  - rich perceptual understanding of the environment,
  - "common-sense" understanding of objects,
  - the ability to make long-term (task-level) plans, and combine them with fine (joint-level) motions.

Motivation

Consider the task of loading a dishwasher…

(from TRI Robotics)







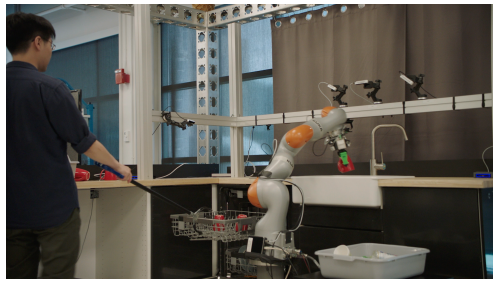










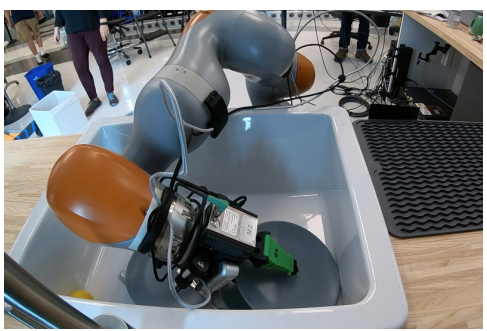


Scene-understanding +

Task-level Planning

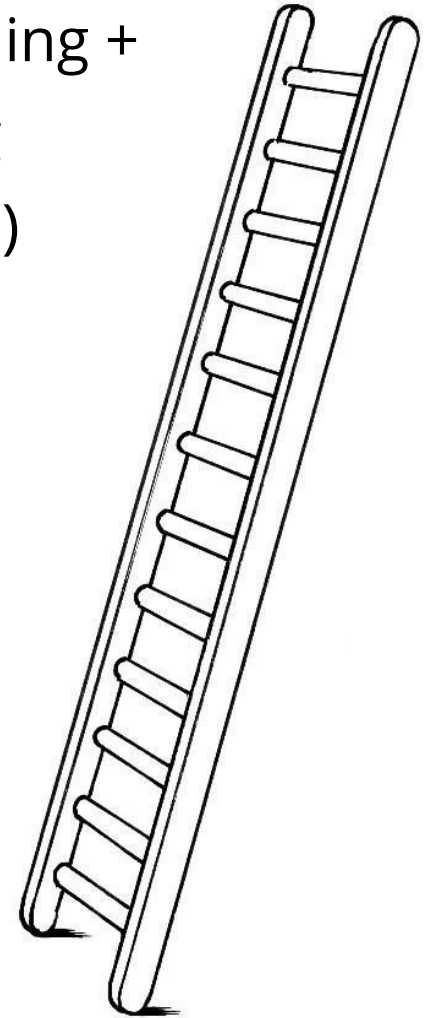
(discrete/symbolic)





Low-level feedback control from cameras / sensors

(continuous time/state/action)



The world is even more "open" when the manipulation system goes *mobile.*


PR2

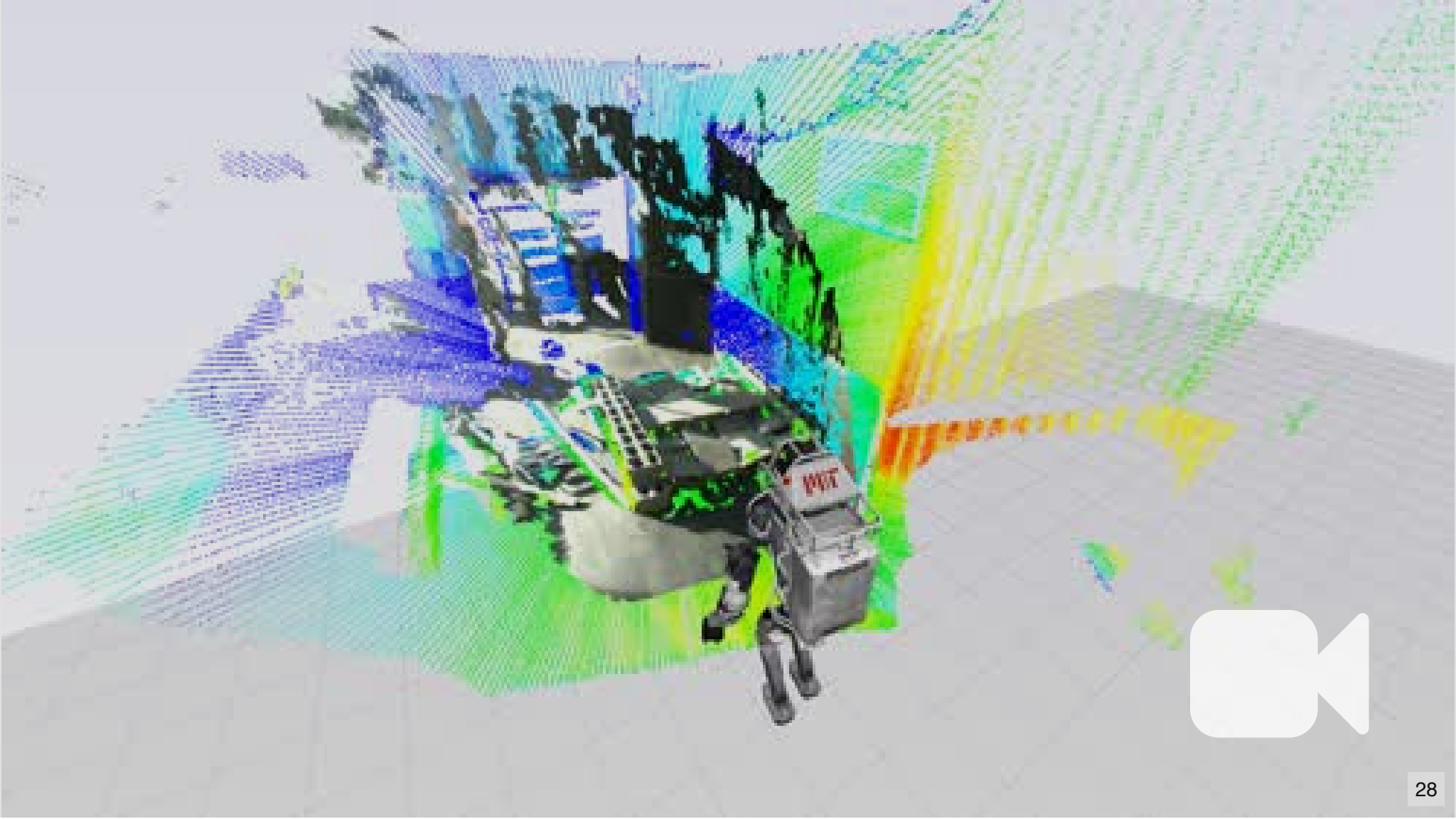


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Manipulation from the *dynamics and controls* perspective…


FAIRPLEX
FAIRPLEX



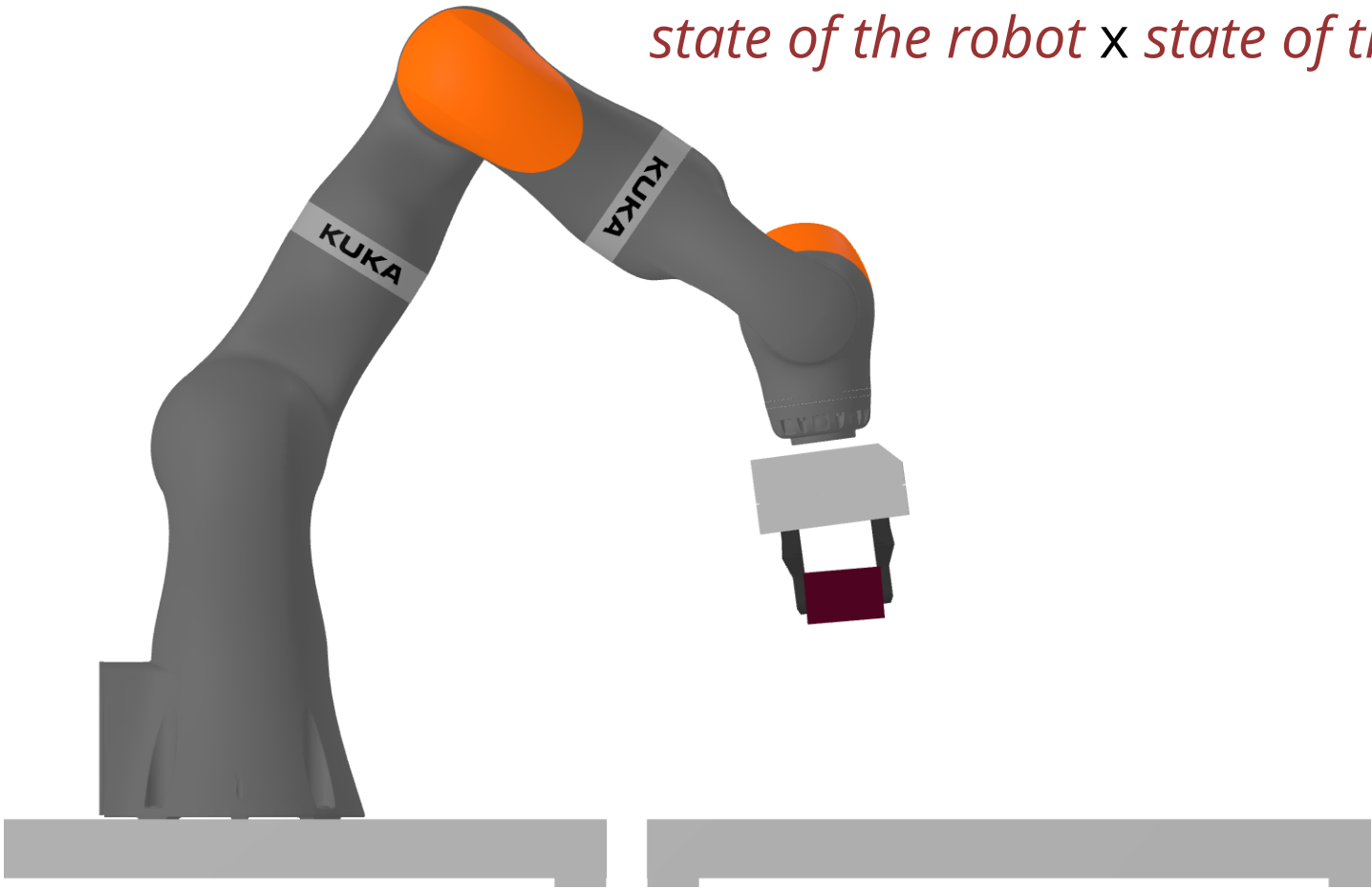





All manipulated objects
in this video have
not been seen before
1x speed

To be clear: we're not just controlling the arm

state of the robot x *state of the environment*

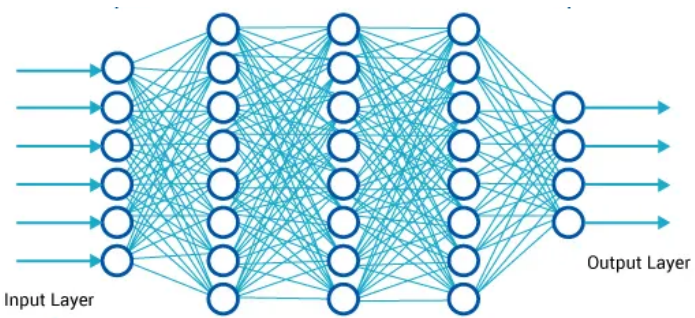


https://www.youtube.com/embed/0LJb66aYtG8?start=111&mute=1&enablejsapi=1&controls=0&rel=0&modestbranding=1

Visuomotor policies

perception network

(often pre-trained)



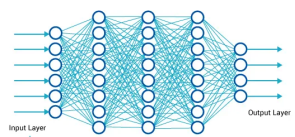
$\mathbb{R}^{640 \times 480 \times 3}$

$\approx \mathbb{R}^{32}$

z

learned state representation

policy network



$\approx \mathbb{R}^{7}$

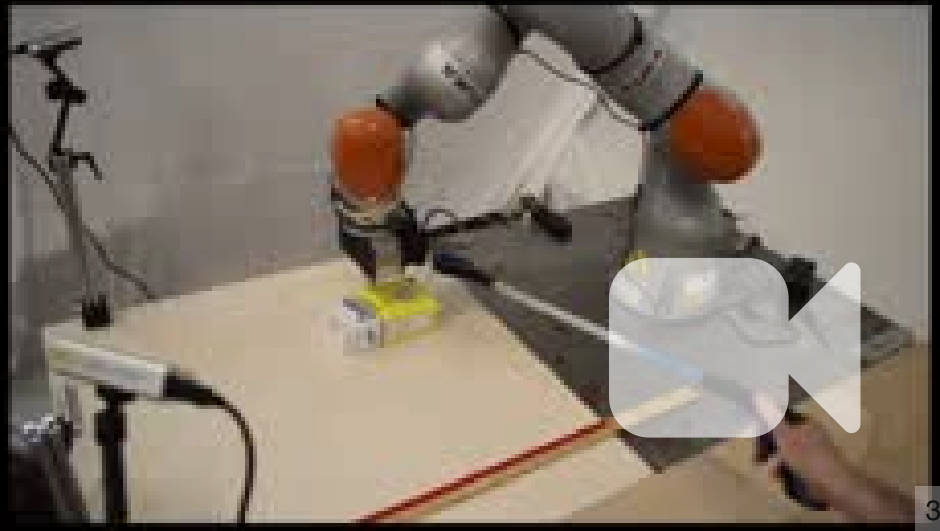
actions

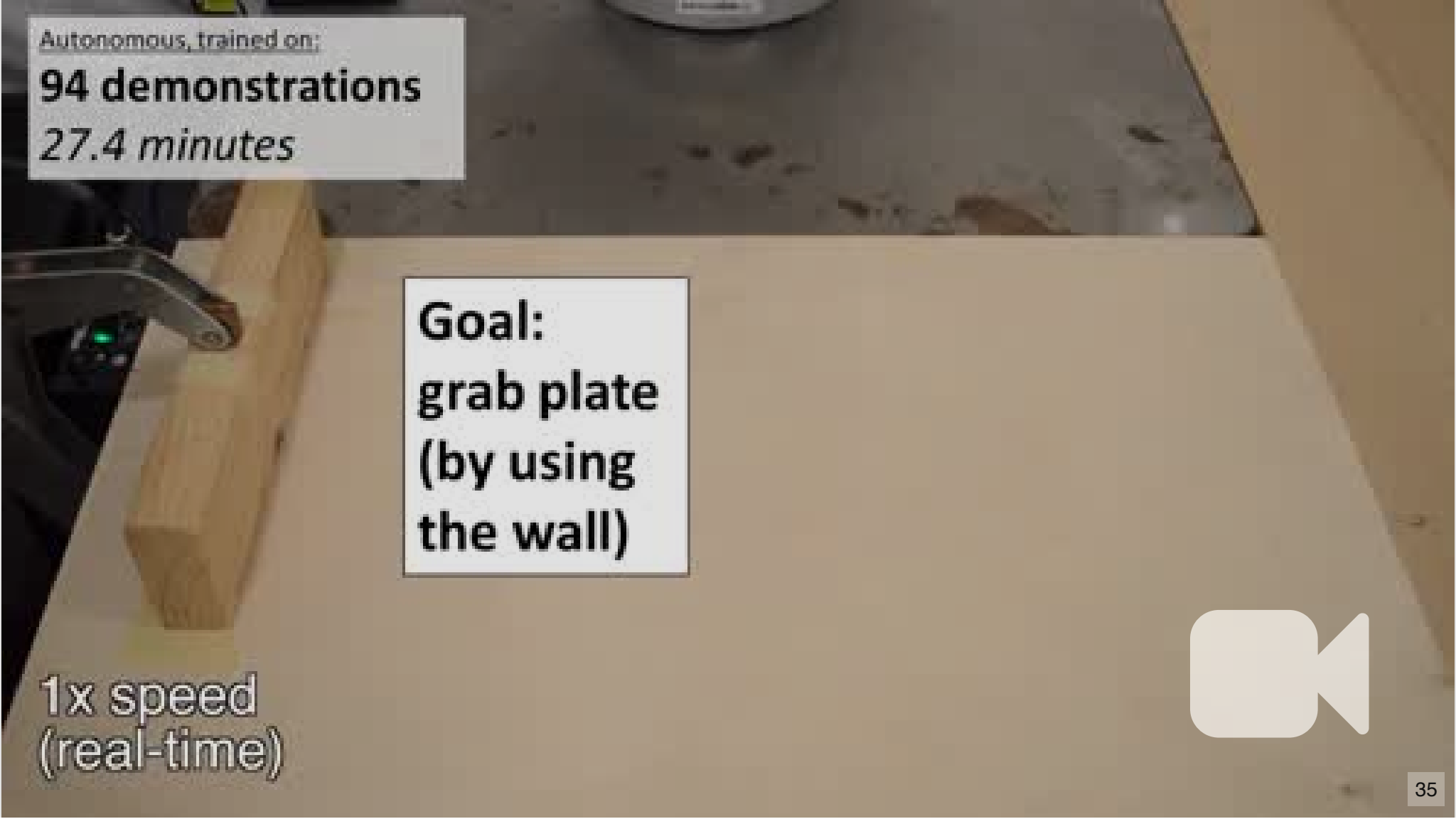
other robot sensors

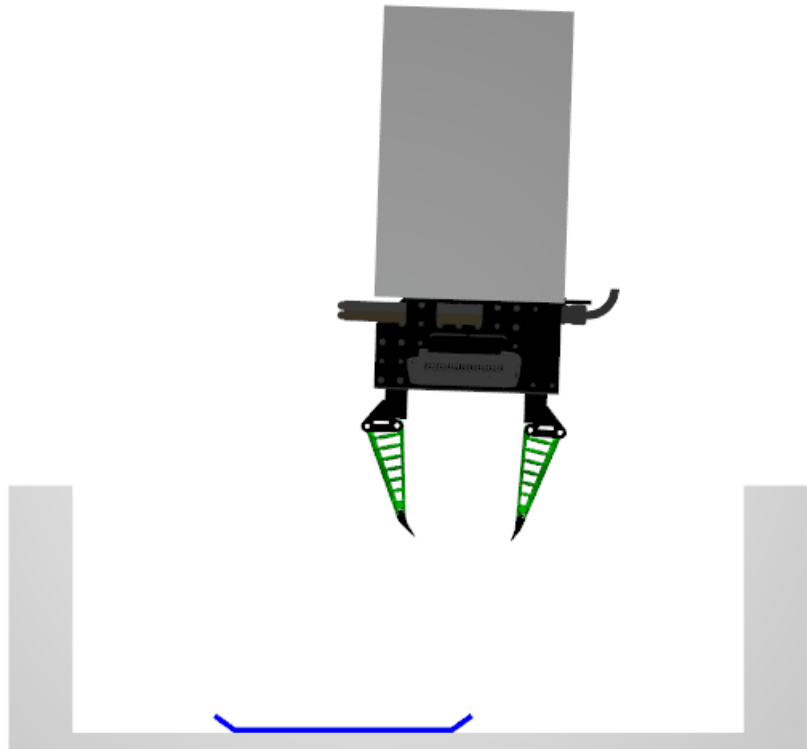
$\approx \mathbb{R}^{13}$

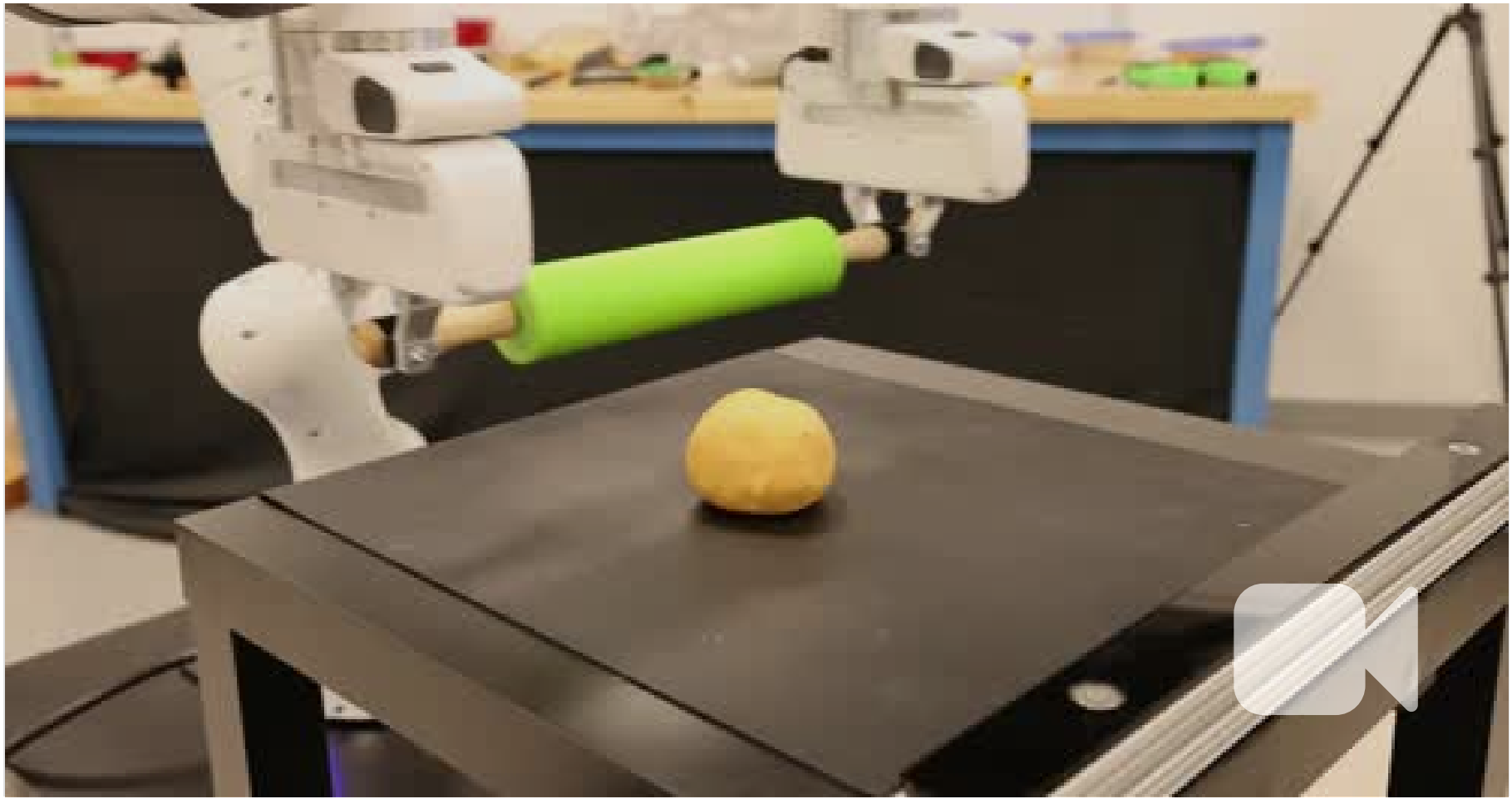
Levine*, Finn*, Darrel, Abbeel, JMLR 2016




Autonomous, trained on:
94 demonstrations
27.4 minutes
Goal:
grab plate
(by using
the wall)
1x speed
(real-time)



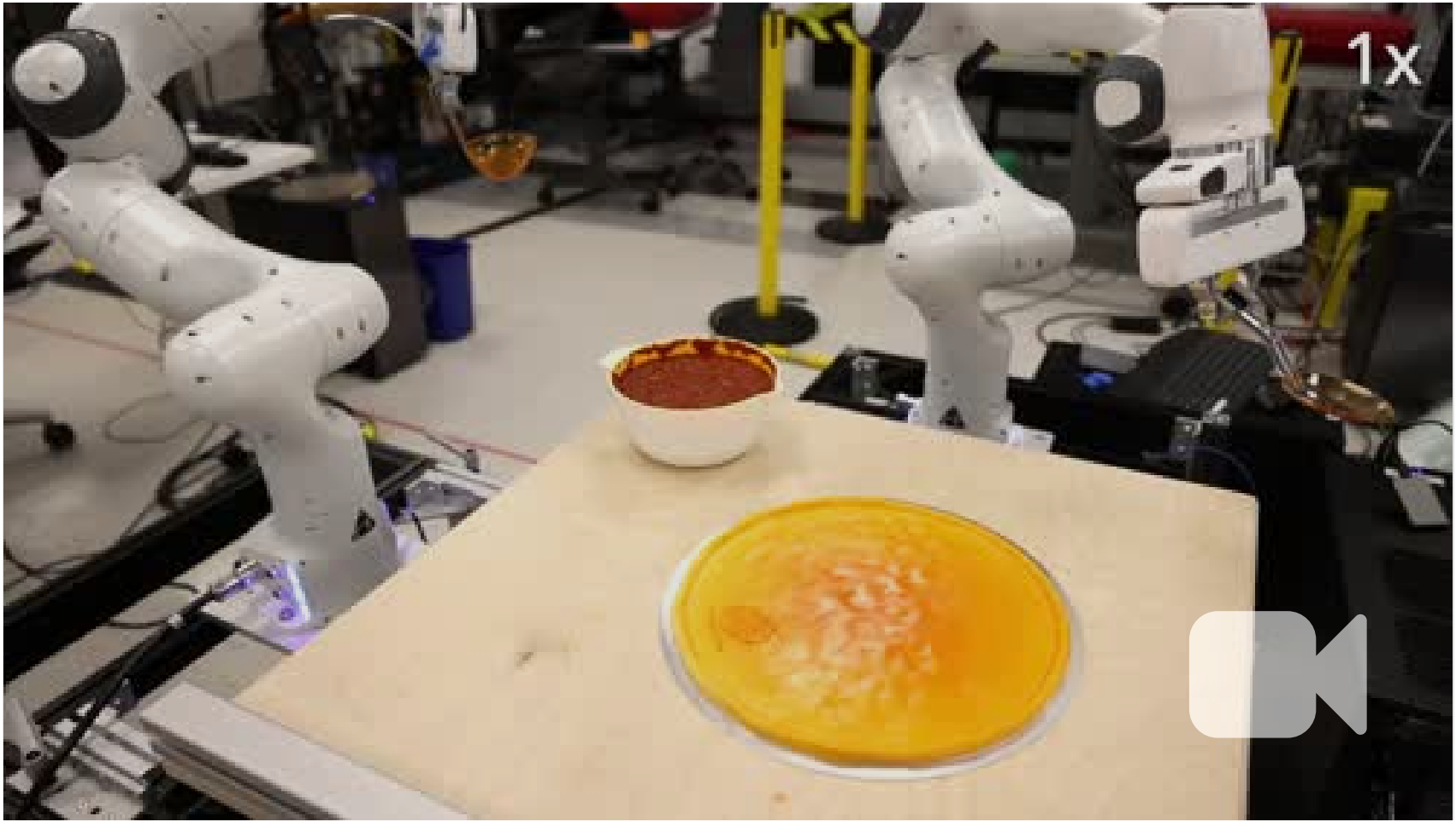


Manipulation is much more than "pick and place"!

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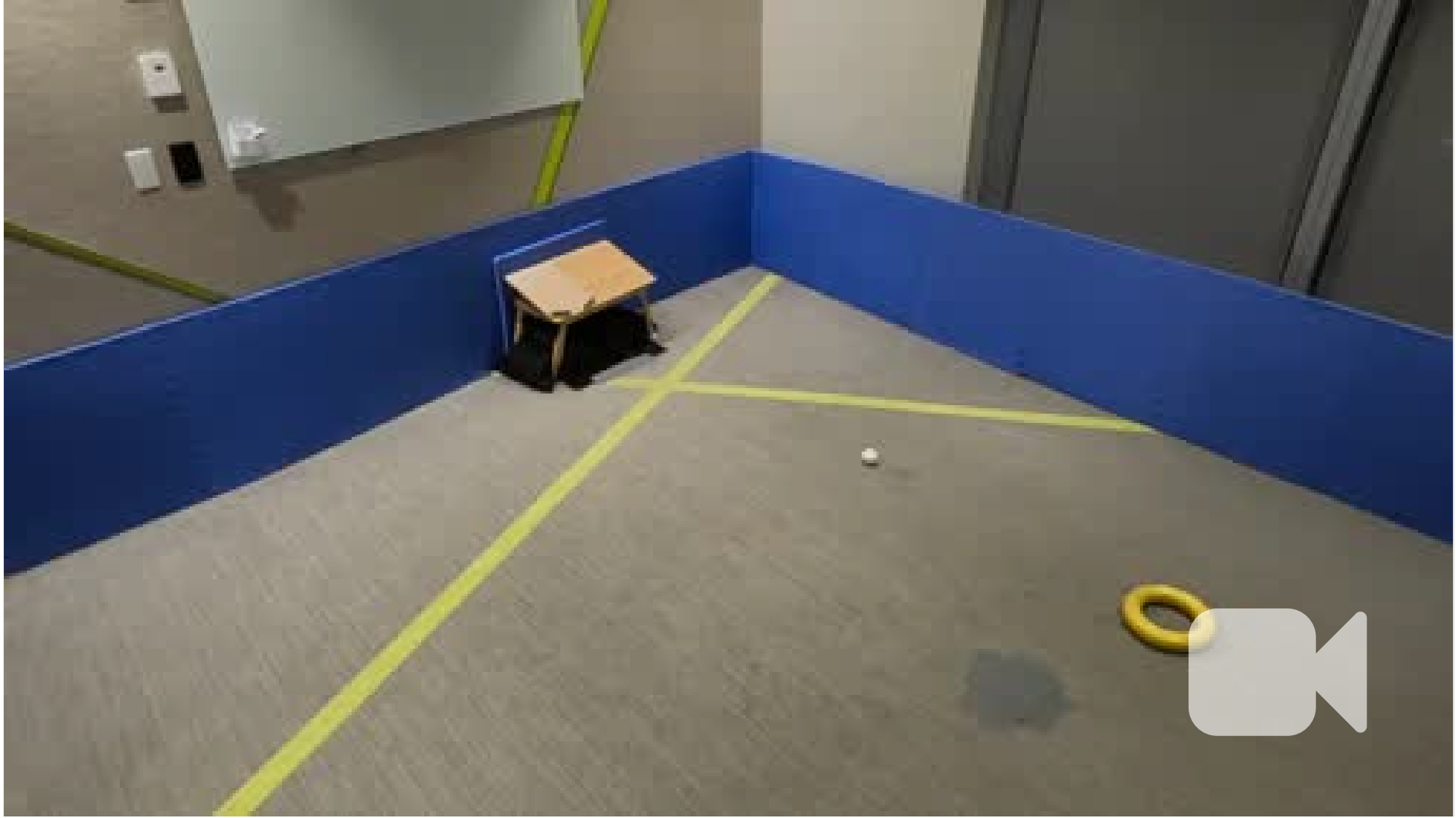


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But even pick and place is challenging. Maybe it's not all visuomotor policies...



Kumar, Silver, McClinton, et al 2024



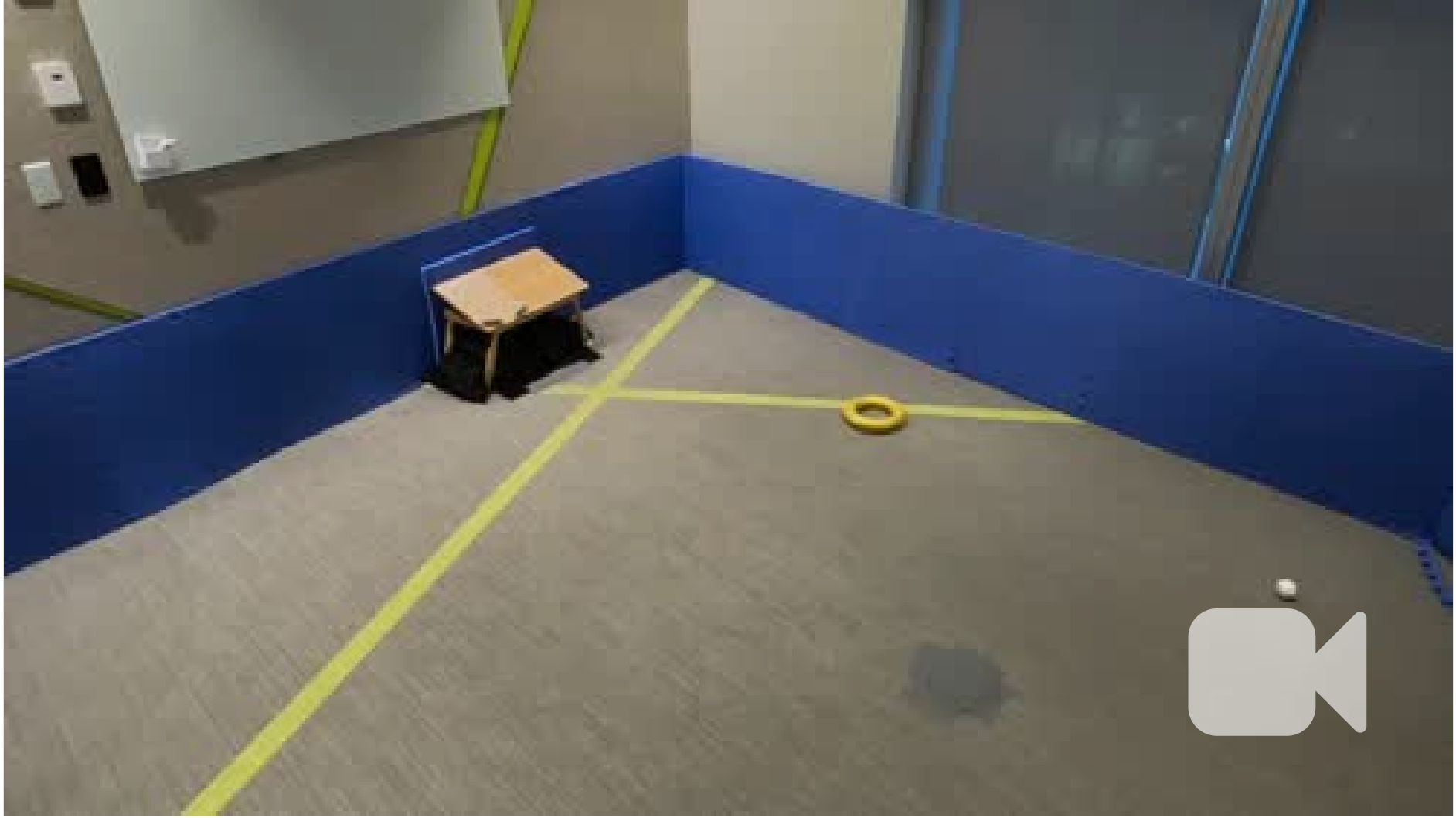
Kumar, Silver, McClinton, et al 2024



Kumar, Silver, McClinton, et al 2024



Kumar, Silver, McClinton, et al 2024



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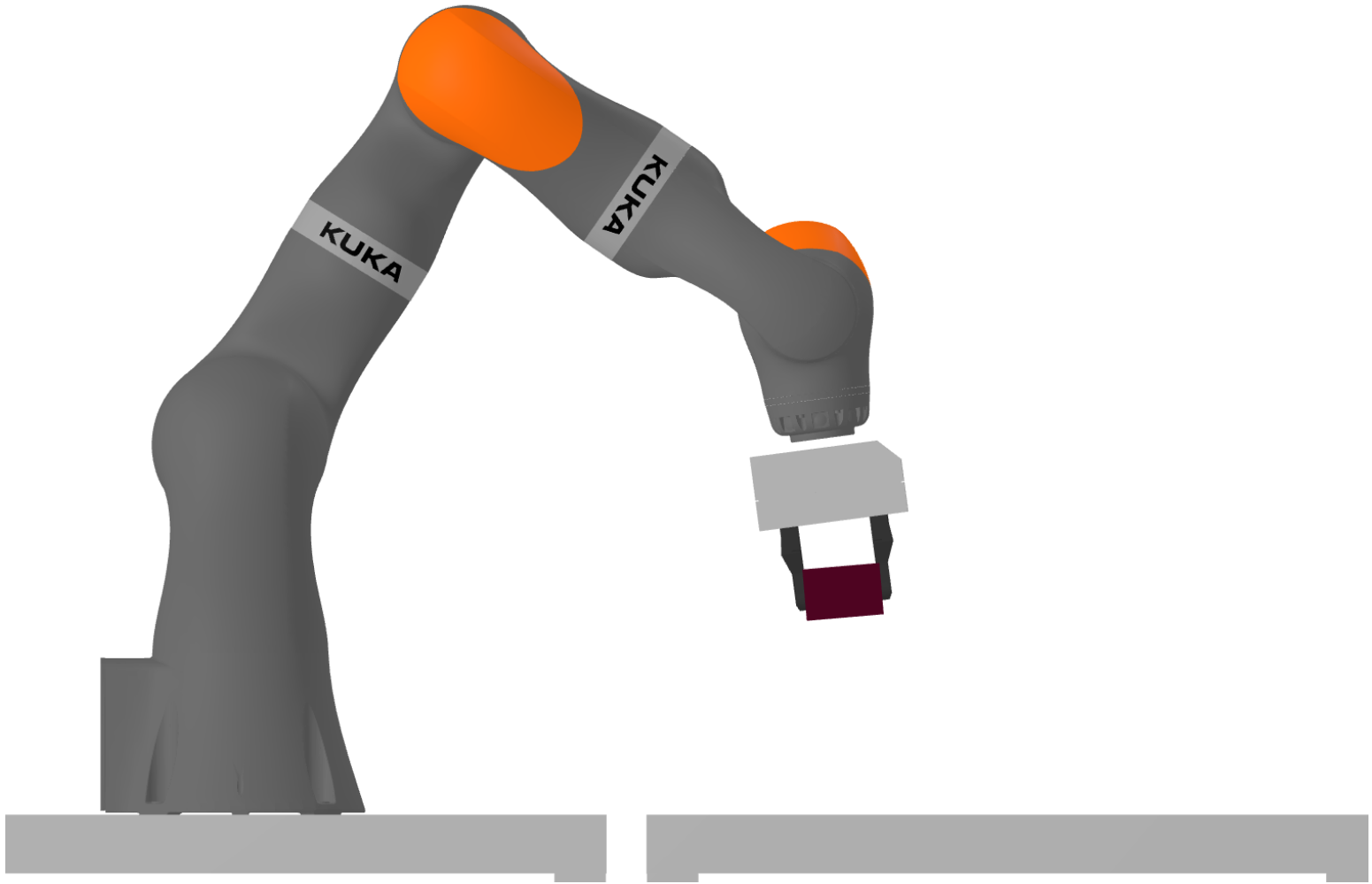
A. Curtis, X. Fang, L. Kaelbling, T. Lozano-Perez, C. Garrett 2022



C. Garrett, C. Paxton, T. Lozano-Perez, L. Kaelbling, D. Fox 2020

The *anatomy* of a modern manipulation system

https://www.ros.org/


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What is Drake?



CORE LIBRARY

Modeling Dynamical Systems	Solving Mathematical Programs	Multibody Kinematics and Dynamics
API | TUTORIAL	API | TUTORIAL	API

drake.mit.edu

Drake Tutorials

For instructions on how to run these tutorial notebooks, please see the README.

Note: We have chosen to host all of our tutorials in Python/Jupyter notebooks, but the `pydrake` API is a thin wrapper over the C++ API. Most code can be ported very easily (in either direction).

For an overview about the philosophy behind Drake, consider reading this blog post.
We have organized the tutorials around the three major components described in that post:

Multibody Dynamics

- Authoring a multibody simulation
- Mathematical programming with MultibodyPlant
- PyPlot animation
- Rendering MultibodyPlant

More multibody tutorials coming soon!

Mathematical Programming

- MathematicalProgram
- Setting solver parameters
- Updating costs and constraints (e.g. for efficient solving of many similar programs)
- Debugging tips
- Linear program
- Quadratic program

Also:
https://github.com/RobotLocomotion/drake-ros
https://github.com/RobotLocomotion/drake-external-examples
...

https://nbviewer.jupyter.org/github/RobotLocomotion/drake/blob/nightly-release/tutorials/dynamical_systems.ipynb

https://manipulation.csail.mit.edu/robot.html#section4

https://manipulation.csail.mit.edu/Fall2024/schedule.html

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